King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 251

Original exam date: 1 October 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int \dfrac{x^3 - 4x^2 - 4x + 20}{x^2 - 5}\, dx =$

(a) $\dfrac{x^2}{2} - 4x - \dfrac{1}{2}\ln|x^2 - 5| + C$

(b) $x^2 + 2x + \dfrac{1}{2}\ln|x^2 - 5| + C$

(c) $x^2 - 2x + \dfrac{1}{2}\ln|x^2 - 5| + C$

(d) $\dfrac{x^2}{2} + 4x + \dfrac{1}{2}\ln|x^2 - 5| + C$

(e) $\dfrac{x^2}{2} - 4x + \dfrac{1}{2}\ln|x^2 - 5| + C$

2. $\int_0^{\ln 2} 2e^{-x}\cosh x\, dx =$

(a) $\dfrac{1}{8} + \ln 2$

(b) $\dfrac{3}{8} + \ln 2$

(c) $\dfrac{3}{8} + \ln 3$

(d) $\dfrac{1}{8} + \ln 3$

(e) $\dfrac{1}{4} + \ln 2$

3. If $\int_{-3}^{3} f(x)\, dx + \int_{3}^{6} f(x)\, dx - \int_{a}^{b} f(x)\, dx = \int_{-1}^{6} f(x)\, dx$, for any integrable function $f(x)$, then $b - a =$

(a) -1

(b) 1

(c) -2

(d) 2

(e) 0

4. The value of c guaranteed by the Mean Value Theorem for integrals for the function $f(x) = \sqrt{x}$ over the interval $[1, 4]$ is

(a) $\frac{49}{9}$

(b) $\frac{196}{9}$

(c) $\frac{144}{81}$

(d) $\frac{196}{81}$

(e) $\frac{144}{9}$

5. The upper sum (in terms of n, the number of subintervals) for the region bounded by the graph of function $f(x) = 5x^2$ and the x-axis on the interval $[0, 1]$ is

(a) $\frac{5}{3} + \frac{5}{3n} + \frac{5}{6n^2}$

(b) $\frac{5}{3} + \frac{5}{2n} + \frac{5}{2n^2}$

(c) $\frac{5}{3} + \frac{5}{2n} + \frac{5}{6n^2}$

(d) $\frac{5}{3} + \frac{5}{2n} + \frac{5}{3n^2}$

(e) $\frac{5}{6} + \frac{5}{2n} + \frac{5}{6n^2}$

6. Use the midpoint rule and 4 rectangles to find an approximation of the area of the region bounded by the graph of the function $f(x) = x^2 + 4x$ and the x-axis over the interval $[0, 4]$. The approximated area is

(a) 57

(b) 54

(c) 55

(d) 56

(e) 53

7. The area of the region bounded by the graphs of the equations $f(x) = \dfrac{1}{9x^2}$, $y = 1$, $x = 1$ and $x = 2$ is

(a) $\dfrac{18}{17}$

(b) $\dfrac{17}{18}$

(c) $\dfrac{16}{15}$

(d) $\dfrac{15}{16}$

(e) 1

8. $\int_{-1}^{1}(1 - |x|)\,dx =$

(a) 8

(b) 1

(c) 4

(d) 2

(e) 3

9. The area of the region bounded by the graphs of the functions $f(x) = x^3 + 2$ and $g(x) = x + 2$ is

(a) 4

(b) $\dfrac{1}{4}$

(c) 2

(d) $\dfrac{1}{2}$

(e) 1

10. $\int \dfrac{dx}{\sqrt{-2x^2+8x+4}} =$

(a) $\sqrt{2}\arcsin\left[\dfrac{\sqrt{6}}{6}(x+2)\right]+C$

(b) $\dfrac{\sqrt{2}}{2}\arcsin(x-2)+C$

(c) $\sqrt{2}\arcsin\left[\dfrac{\sqrt{6}}{6}(x-2)\right]+C$

(d) $\dfrac{\sqrt{2}}{2}\arcsin\left[\dfrac{\sqrt{6}}{6}(x-2)\right]+C$

(e) $\dfrac{\sqrt{2}}{2}\arcsin\left[\dfrac{\sqrt{6}}{6}(x+2)\right]+C$

11. $\int \dfrac{x^3}{\sqrt{1+x^4}}\,dx =$

(a) $\sqrt{1+x^4}+C$

(b) $\dfrac{1}{2}\ln(1+x^4)+C$

(c) $\dfrac{1}{4}\sqrt{1+x^4}+C$

(d) $\dfrac{1}{2}\sqrt{1+x^4}+C$

(e) $2\sqrt{1+x^4}+C$

12. Let the velocity function $v(t)=t^2-t-12$ in feet per second, be given for a particle moving along a straight line, where t is the time in seconds. The total distance (in feet) that the particle travels over the interval $[0,5]$ is equal to

(a) $\dfrac{71}{2}$

(b) $\dfrac{73}{2}$

(c) $\dfrac{77}{2}$

(d) $\dfrac{79}{2}$

(e) $\dfrac{75}{2}$

13. $\int \dfrac{\sec(2\ln t) + \tan(2\ln t)}{t}\, dt =$

(a) $-\ln|1 - \sin(2\ln t)| + C$

(b) $-\dfrac{1}{2}\ln|1 - \sin(2\ln t)| + C$

(c) $\ln|1 - \sin(2\ln t)| + C$

(d) $\dfrac{1}{4}\ln|1 - \sin(2\ln t)| + C$

(e) $\dfrac{1}{2}\ln|1 - \sin(2\ln t)| + C$

14. If $F(x) = \int_0^{x^3} \sin t^2\, dt$, then $\sqrt[3]{\dfrac{6}{\pi}}\, F'\left(\sqrt[6]{\dfrac{\pi}{6}}\right) =$

(a) 2

(b) 3

(c) 1

(d) $\dfrac{3}{2}$

(e) $\dfrac{1}{2}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Divide: $x-4+\frac{x}{x^2-5}$, giving $\frac{x^2}{2}-4x+\frac{1}{2}\ln\lvert x^2-5\rvert+C$.
2	**(b)**	Integrand $=1+e^{-2x}$; value $=\ln 2+\frac{3}{8}$.
3	**(d)**	Additivity gives $\int_a^b f=\int_{-3}^{-1} f$, so $a=-3,\ b=-1,\ b-a=2$.
4	**(d)**	Average $=\frac{14}{9}=\sqrt{c}$, so $c=\frac{196}{81}$.
5	**(c)**	Right endpoints: $\frac{5}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}=\frac{5}{3}+\frac{5}{2n}+\frac{5}{6n^2}$.
6	**(e)**	Midpoints $0.5, 1.5, 2.5, 3.5$: $2.25+8.25+16.25+26.25=53$.
7	**(b)**	$\int_1^2\left(1-\frac{1}{9x^2}\right)dx=\frac{17}{18}$.
8	**(b)**	Triangle of base 2 and height 1: area $=1$.
9	**(d)**	Intersections $-1, 0, 1$; $A=\frac{1}{4}+\frac{1}{4}=\frac{1}{2}$.
10	**(d)**	$-2x^2+8x+4=2[6-(x-2)^2]$, so $\frac{\sqrt{2}}{2}\arcsin\frac{\sqrt{6}}{6}(x-2)+C$.
11	**(d)**	$u=1+x^4$: $\frac{1}{4}\int u^{-1/2}du=\frac{1}{2}\sqrt{1+x^4}+C$.
12	**(c)**	$\lvert v\rvert$ split at $t=4$: $\frac{104}{3}+\frac{23}{6}=\frac{77}{2}$.
13	**(b)**	$u=2\ln t$, $\sec u+\tan u=\frac{\cos u}{1-\sin u}$: $-\frac{1}{2}\ln\lvert 1-\sin(2\ln t)\rvert+C$.
14	**(d)**	$F'(x)=3x^2\sin x^6$; at the given x the product is $\frac{3}{2}$.